Choosing an analysis method

Concepts of Biostatistics and Data Science: a two-day crash course with practical analysis in R

Start from the **question**, then the **outcome type**, then the **design** (independent or paired), then the number of **groups or predictors**. The table suggests a usual first choice and what to report; it is a guide for thinking, not an automatic rule. The decision diagrams D0 to D6 show the same choices visually, and the cheat sheet gives assumptions, R code and common mistakes for each method. *Grey italic entries* are beyond this course and are routes for further learning.

Question	Outcome	Independent observations	Paired observations	Report	R, course example
Describe (diagram D1)					
What do the data look like?	Categorical	Counts and % (state the denominator); bar chart		n, %	`count(hearts, district)`
	Continuous	Mean (SD) if symmetric; median (IQR) if skewed; histogram, boxplot		n, missing, summary	`summarise(hearts, mean(sbp_0), sd(sbp_0))`
Compare two groups (D2, D3)					
Do two groups differ?	Continuous	Welch t-test (default); Student's t only with equal spread	Paired t-test on the differences	Mean difference, 95% CI	`t.test(sbp_0 ~ sex, data = hearts)`
	Continuous, positive, right-skewed	t-test on log scale	Paired t-test on log values	Ratio of geometric means, CI	`t.test(log(crp) ~ smoker, data = hearts)`
	Ordinal, or ranks suit the question	Wilcoxon rank-sum (Mann-Whitney)	Wilcoxon signed-rank	Medians (IQR); Hodges-Lehmann shift, CI	`wilcox.test(crp ~ smoker, data = hearts, conf.int = TRUE)`
	Binary	Chi-square; Fisher's exact when expected counts are small	McNemar's test	Proportions; RD, RR or PR, OR with CI	`chisq.test(tab)`; `fisher.test(tab_ae)`; `mcnemar.test(tab_paired)`
	Count with follow-up time	Poisson regression with offset	*conditional models*	Rate ratio (IRR), CI	`glm(clinic_visits ~ programme + offset(log(followup_months)), family = poisson, data = hearts)`
	Time-to-event	*Kaplan-Meier, log-rank test*		*Hazard ratio*	*survival package*
Compare three or more groups (D2, D3)					
Do several groups differ?	Continuous	Welch ANOVA or classical ANOVA; then adjusted pairwise comparisons if needed	*Repeated-measures models, mixed models*	Group means with CIs; adjusted pairwise differences	`oneway.test(sbp_0 ~ district, data = hearts)`; `TukeyHSD(aov(...))`
	Ordinal, or ranks	Kruskal-Wallis	*Friedman test*	Medians (IQR) per group	`kruskal.test(crp ~ district, data = hearts)`

Question	Outcome	Independent observations	Paired observations	Report	R, course example
	Categorical	Chi-square for an r x c table; Fisher when counts are small	*Cochran's Q*	Row % per group	`chisq.test(table(hearts$district, hearts$high_bp_0))`
Assess an association between two variables (D3, D4)					
Are two variables related?	Two continuous	Scatterplot; Pearson r (linear) or Spearman rho (monotonic); simple linear regression for the size of change	*Repeated or clustered: mixed models*	r with CI; slope with CI	`cor.test(~ sbp_0 + age, data = hearts)`; `lm(sbp_0 ~ age10, data = hearts)`
	Two categorical	Contingency table, chi-square or Fisher	McNemar (same binary variable twice)	RD, RR, PR or OR by design	`prop.table(tab, margin = 1)`
Estimate an association adjusted for other variables (D4)					
What is the association, holding confounders constant?	Continuous	Multiple linear regression	*Mixed models, GEE*	Adjusted slope, CI	`lm(sbp_0 ~ activity_hr + age + sex, data = hearts)`
	Binary	Logistic regression	*Conditional logistic regression (matched), GEE*	Adjusted OR (aOR), CI	`glm(high_bp_0 ~ smoker + age10 + sex, family = binomial, data = hearts)`
	Count	Poisson regression with offset; quasi-Poisson or negative binomial if overdispersed	*Mixed models, GEE*	Adjusted IRR, CI	`glm(... + offset(log(followup_months)), family = poisson, ...)`; `MASS::glm.nb()`
	Ordinal	*Proportional odds regression*		*Cumulative OR*	*MASS::polr*
	Nominal, more than 2 categories	*Multinomial logistic regression*		*Relative risk ratios*	*nnet::multinom*
	Time-to-event	*Cox proportional hazards regression*		*Hazard ratio*	*survival::coxph*
Predict an outcome for new people (D4)					
How well can we predict?	Any	The regression model for the outcome type, fitted on training data and judged on test data; no leakage		Prediction error on test data (e.g. RMSE)	`train/test split with sample()`; `predict(fit, newdata = test)`
Discover groups (D5)					
Are there subgroups?	None (no outcome)	Scale variables in different units; k-means or hierarchical clustering; check k, stability and meaning		Cluster sizes and profiles; how k was chosen	`kmeans(scale(x), centers = 3, nstart = 25)`; `hclust(dist(z), method = "ward.D2")`

When basic independent-observation methods are inappropriate (diagram D6)

Structure	Example	Problem if ignored	What to do
Paired	Same people before and after; matched case and control	Independent-groups tests give wrong (usually too wide) CIs	Analyse differences: paired t-test, Wilcoxon signed-rank, McNemar (covered in the course)
Repeated measures	SBP at 0, 3, 6 and 12 months	Separate tests at each time point inflate false positives; people counted as several independent observations	One summary per person (change, slope, mean) analysed with basic methods; *mixed-effects models or GEE*
Clustered	Patients within clinics; people within households or districts	Standard errors too small, CIs too narrow, p-values too small	*Mixed-effects (multilevel) models, GEE, cluster-robust standard errors*; with few clusters, analyse cluster summaries
Time-to-event with censoring	Time to diagnosis; some people not diagnosed by the end of follow-up	Dropping censored people or using event yes/no biases results when follow-up differs	*Kaplan-Meier, log-rank test, Cox regression*
Different follow-up times for counts	Clinic visits over 2 to 12 months	Raw counts compare unequal exposure	Poisson regression with `offset(log(followup_months))` (covered)

Principles that matter more than any flowchart

- **Sample size alone does not decide between a t-test and a rank-based test.** With small samples, look carefully at the data (strong skew, outliers, ordinal scale) and choose the summary that answers the question. Rank-based tests are not a safety net for small samples: they also have assumptions and can have little power.
- **Do not choose a test from a normality-test p-value.** In large samples such tests flag trivial departures; in small samples they miss important ones. Plots of the data and residuals are more informative. For t-tests, ANOVA and regression what matters is the distribution of the sample means or residuals, not of the raw outcome.
- **Rank-based tests are not assumption-free and not, in general, tests of medians.** Wilcoxon and Kruskal-Wallis assume independent observations and compare whole distributions through ranks.
- **A significant overall test (ANOVA, Kruskal-Wallis, r x c chi-square) does not identify which groups differ.** Use adjusted pairwise comparisons or look at the estimates.
- **Welch's versions are sensible defaults** for two-group and several-group comparisons of means, because equal variances are rarely known.
- **Independence comes from the design**, not from a test. Paired, repeated and clustered data need methods that respect their structure.
- **Report effect estimates with 95% CIs.** Statistical significance does not establish practical importance or causation, and a non-significant result does not show that there is no effect.
- **The study design determines which measures can be interpreted:** risks and RR from cohorts and trials, prevalences and PR from cross-sectional studies, only the OR from case-control studies.
- **Choose adjustment variables from subject knowledge**: adjust for confounders, not mediators.



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